Object Oriented Programming Lab

Lab 10

Instructions

Work on this lab individually. You can use your books, notes, handouts etc but you are not allowed to borrow anything from your peer student.

Write down the lab number, task number, your name and roll number at the start of each task code using comments.

For Example

```
/***********************************
        Lab:      xx
        Task:     yy
        Developer:
                  Roll #:   XXXYYYZ000
                  Name:     XYZ
***********************************/
```

What you have to do

Program the following tasks in C++, compile and execute them.

Task

You are required to design a Shape measurement system using inheritance and polymorphism.
Create a base class Shape with following attributes and functions:

- Data members (protected):
  - string name
  - String type
- Member functions:
  - A constructor to initialize data members
  - virtual double getArea() {}
  - virtual double getVolume() {}
  - A function void showInfo() to display shape name and type

Create Three Derived Classes. Each class must inherit publicly from Shape and override **getArea()** and **getVolume().**

Triangle (2D)

- Additional attribute: `double base, double height`
- `getArea() returns:` 0.5 * base * height
- `getVolume() returns: 0 (no volume of 2D shape)`

Cube (3D)

- Additional attribute: `double` side
- `getArea() returns:` 6 * side * side
- `getVolume() returns:` side * side * side

Cylinder (3D)

- Additional attribute: `double` radius, double height
- `getArea() returns:` 2 * π * radius * (radius + height)
- `getVolume() returns:` π * radius * radius * height

In main()

1. Declare a parent class reference **Shape *sh;**
2. Create objects of child classes, assign each object to the parent-class reference and make a call to `getArea()` and `getVolume()`.Observe that the correct version of getArea() and getVolume() executes for each object.

☺ ☺ ☺ **BEST OF LUCK** ☺ ☺ ☺